Robert Todd

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EDUCATION

Carnegie Mellon University — *M.S. Computer Science (Software Engineering)* 2005 - 2007

University of Texas at Austin — *B.S. Computer Science* 2001 - 2005

EXPERIENCE

Stripe — *Principal Quality Assurance Engineer* 2021 - Present

- Architected and deployed enterprise-grade test automation platform reducing payment processing regression detection time from 8 hours to 12 minutes across 15+ payment integrations affecting $500B+ annual transaction volume.
- Led organization-wide quality transformation initiative establishing QA best practices, test pyramid standards, and metrics frameworks resulting in 67% reduction in production defects and $12M annual savings in incident response costs.
- Pioneered AI-powered visual regression testing system using machine learning models to detect UI anomalies across 50+ supported locales, reducing manual review cycles by 85% and enabling real-time deployment to 400M+ users.

Meta — *Staff Quality Engineer - Ads Platform* 2018 - 2021

- Designed and implemented distributed load testing infrastructure leveraging Kubernetes clusters handling 50K+ concurrent requests, validating ad delivery system performance under peak load and preventing $200M+ daily revenue impact from outages.
- Established chaos engineering practice across Ads Platform team, designing 300+ chaos experiments that identified and prevented 47 critical reliability issues before customer impact, improving system MTBF by 156%.
- Published peer-reviewed paper 'Continuous Chaos: Scaling Resilience Testing in High-Frequency Trading Systems' presented at ICSE 2020, cited 340+ times, influencing QA practices at 8+ Fortune 500 companies.

Jane Street — *Principal Engineer, Quantitative Testing & Validation* 2015 - 2018

- Architected statistical testing framework for high-frequency trading algorithms combining Monte Carlo simulations with hypothesis testing, validating mathematical models for 200+ trading strategies processing $40B+ daily market volume.
- Reduced mean time to detection for market microstructure anomalies from 4.2 seconds to 340 milliseconds through custom test harness and telemetry optimization, protecting proprietary strategies from adversarial detection.
- Mentored 22 junior engineers on formal verification methods and probabilistic testing techniques; three mentees promoted to senior engineer roles within 18 months.

Uber — *Senior Quality Architect* 2012 - 2015

- Designed end-to-end test orchestration platform automating 85,000+ test cases across mobile (iOS/Android), backend services, and driver/consumer app stacks, reducing regression cycle time by 73% and enabling 8+ daily deployments.
- Led development of location-based testing framework simulating real-world GPS conditions and edge cases across 65+ countries, identifying critical navigation bugs affecting 4M+ daily active riders and preventing $50M+ customer acquisition loss.
- Established QA guild with representation from 12 engineering teams; drove adoption of contract testing for 150+ microservices, eliminating 94% of cross-service integration failures and reducing incident response time by 58%.

Qualcomm — *Distinguished Quality Engineer* 2007 - 2012

- Led quality assurance for flagship 5G modem chipset validation involving 2000+ test cases across physical layer, MAC layer, and system integration across 300+ device configurations, achieving 99.94% field reliability.
- Pioneered industry-first regression test compression algorithm reducing modem validation cycle from 96 hours to 14 hours while maintaining 99.8% defect detection sensitivity, accelerating time-to-market by 6 weeks for $3.2B product line.
- Presented keynote 'Scaling QA in the Era of Connected Devices' at QA Summit 2011 attended by 1200+ engineers; insights adopted by 25+ semiconductor companies for hardware validation processes.

EXTRACURRICULAR

IEEE Software Testing Technical Council — *Vice Chair* 2019 - Present

- Oversee technical roadmap and standards development for AI-assisted testing across 15,000+ members in 80+ countries.
- Organized annual Software Testing Summit attracting 2500+ practitioners with keynotes from industry leaders at Google, Microsoft, and Amazon.

ACM SIGSOFT — *Program Committee Member* 2016 - Present

- Reviewed 200+ research submissions on software testing, quality assurance, and formal verification for ICSE, FSE, and ASE conferences.
- Selected as distinguished reviewer for top 5% of submissions with exceptional technical rigor and novelty.

TECHNICAL SKILLS

Languages – Python, Java, JavaScript, SQL, Bash, Groovy

Technologies – Test Automation Frameworks (Selenium, Cypress, Playwright), API Testing (REST, gRPC), Load Testing (JMeter, Gatling), CI/CD Pipelines, Test Data Management, Defect Lifecycle Management

Tools – TestRail, Jira, Jenkins, Docker, Kubernetes, AWS (EC2, Lambda, RDS), GitLab CI/CD, ELK Stack, Grafana, Datadog